PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

GeoDefence: Interactive Threat Mapping and Monitoring System

CSS7000 INTERNSHIP REPORT

Submitted by

FELIX K S - 20241CAI0086

Under the guidance of,

Dr.Srinivasan T R

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence and Machine Learning)

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

PRESIDENCY SCHOOL OF
ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING
BONAFIDE CERTIFICATE

Certified that this report **GeoDefence: Interactive Threat Mapping and Monitoring System** is a bonafide work of **FELIX K S - 20241CAI0086** , who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING , during 2026-2027.

Dr Srinivasan T R
Professor
Internship Guide
PSAIAC
Presidency University

Mr.Anandan
Assistant Professor
Program Internship Coordinator
PSAIAC
Presidency University

Dr. Geetha Arjunan
Associate Professor
School Internship Coordinator
PSAIAC
Presidency University

Dr. Zafar Ali Khan N
Professor & Head of the Department
PSAIAC
Presidency University

Dr. Shakkeera L
Dean
PSAIAC
Presidency University

Sl.No	Name of the Examiner	Signature
1.	Mr.John Bennet J	

PRESIDENCY SCHOOL OF
ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING
DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence And Machine Learning), at Presidency University, Bengaluru, named FELIX K S hereby declare that the internship work titled **GeoDefence: Interactive Threat Mapping and Monitoring System** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2026-2027. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

FELIX K S	20241CAI0086	Signature

PLACE: BENGALURU

DATE: 20/8/2026

INTERNSHIP COMPLETION CERTIFICATE

ATWIC Research & Development Pvt Ltd

ATWIC R&D
INNOVATIVE CREATIONS

CORP Office:
ATWIC Research and Development Pvt Ltd
Building No: 16/1108, Green garden Road
CSEZ PO,Kakkanad, KOCHI - 682037, Kerala, INDIA

REG Office:
ATWIC Research and Development Pvt Ltd
SY NO.483/5, Palaparambil Veedu, Near Elambathkaav,
Rajagiri Valley P.O, KOCHI - 682039, Kerala, INDIA

ARD-2026-HR-034
AUG 5th 2026

INTERNSHIP CERTIFICATE

This is to certify that **FELIX K S** (Reg. No.: **20241CAI0086**), a **Semester IV** student of the **Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), Presidency University, Bangalore**, has successfully completed an internship at **ATWIC Research and Development Pvt. Ltd.** in the **AI/ML Department** from **20/07/2026 to 04/08/2026**.

During the internship, he worked on the project titled **"GeoDefence: Interactive Threat Mapping and Monitoring System."**

Throughout the internship period, he demonstrated dedication, enthusiasm, and a strong willingness to learn. His performance and conduct were found to be satisfactory.

We appreciate his efforts and wish him every success in his future academic and professional endeavors.

Yours, Faithfully,

ATWIC Research and Development Pvt Ltd.

For ATWIC RESEARCH & DEVELOPMENT PVT. LTD.
Managing Director

Rahul Sivan
Managing Director - CEO
ATWIC Research and Development Pvt Ltd.

CIN	: U73100KL2015PTC038532	Mail	: info@atwic.com
GST	: 32AANCA4796K1Z6	Web	: www.atwic.com
PAN	: AANCA4796K	Ph	: +91-484-2100499
UAN	: KL02D0002087	Mob	: 09847912899 / 09447083169

Page 1 of 1

ACKNOWLEDGEMENT

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved Chancellor,Vice Chancellor , Pro-Vice Chancellor, and Registrar for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide Dr.Srinivasan T R , Professor Presidency School of Artificial Intelligence & Advanced Computing, Presidency University, for his moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to Dr.Zafar Ali Khan N, Professor, Head of the Department, Presidency School Of Artificial Intelligence & Advanced Computing, Presidency University, for his mentorship and encouragement.

I express my cordial thanks to Dr. Shakkeera L, Dean, Presidency School Of Artificial Intelligence & Advanced Computing, Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

We are grateful to Dr. Geetha Arjunan,Associate Professor, School Internship Coordinator, Mr.Anandan B, Assistant Professor, Program Internship Coordinator , Presidency School Of Artificial Intelligence & Advanced Computing,Presidency University for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School Of Artificial Intelligence & Advanced Computing and also staff from other departments who have extended their valuable help and cooperation.

FELIX K S

TABLE OF CONTENTS

ABSTRACT

GeoDefence: Interactive Threat Mapping and Monitoring System is a prototype developed to make the process of marking, monitoring, and managing location-based threats more simple and organized. The main idea behind the project is to provide users with an interactive map interface where important locations or potential threat areas can be identified and recorded easily. Instead of managing location information manually, the system brings geographical visualization and user interaction together in a single application.

The project is developed using Python and PyQt6 for the desktop application, while Leaflet is used to provide an interactive map interface. The current version of GeoDefence uses online map services to display geographical information. The application provides multiple map views, allowing the user to interact with and explore different geographical areas more effectively.

A major feature of the system is the ability to manually mark locations directly on the map. When a user identifies a location of interest, a marker can be placed at that point. The user can then provide additional information about the marked location, such as a threat level and a description. The threat level can be categorized into different levels, including High, Medium, and Low, helping users quickly understand the importance of each marked location.

The application is designed to be interactive and flexible. Users can edit the information associated with a marked location, change its threat level, move the marker to a new position, or delete an individual detection when it is no longer required. A Clear All option is also provided to remove all marked locations when necessary. These features allow the user to manage the map according to changing requirements without needing technical knowledge or modifications to the source code.

Another important aspect of GeoDefence is the management of user-generated information. The system is designed to support the saving and persistence of details entered by the user, such as geographical coordinates, threat classifications, and descriptions. This provides a foundation for integrating a dedicated backend or database system for long-term storage and retrieval of recorded information.

The current project focuses on creating a stable and functional prototype that

demonstrates the core concept of interactive threat mapping. It is not intended to be presented as a complete operational defence system. However, the prototype provides a strong base for future improvements. Possible future developments include offline map integration, database connectivity, centralized data management, real-time synchronization, GPS support, additional geographical layers, and automated threat identification using artificial intelligence and machine learning techniques.

In conclusion, GeoDefence: Interactive Threat Mapping and Monitoring System demonstrates how Python-based desktop applications and interactive web-mapping technologies can be combined to create a practical location-monitoring system. The project enables users to visually identify locations, record relevant information, classify potential threats, and manage marked points through a simple and user-friendly interface. The prototype serves as a foundation for further development into a more advanced geospatial monitoring and situational-awareness system.

List of Figures

1. Introduction – GeoDefense Concept and Storyline

1.1 Introduction

In situations where geographical information is important, identifying, recording, and monitoring locations efficiently can make a significant difference in decision-making and overall situational awareness. Information about a particular location may be collected from observations, reports, field personnel, sensors, or other available sources. However, managing this information without a proper geographical interface can make it difficult to visualize locations, understand their significance, and monitor multiple points effectively.

When location-related information is recorded only as text or separate data entries, it becomes challenging to understand the geographical relationship between different locations. A visual map-based system can help organize this information in a more structured and understandable manner. It allows users to view multiple locations on a single interface, access relevant information about each point, and classify them based on their importance or priority.

Therefore, an interactive geographic monitoring system can provide a practical solution for managing location-based information. By combining an interactive map with features such as location marking, threat classification, descriptions, and monitoring statistics, users can obtain a clearer overview of the recorded information. This approach improves the visualization and organization of geographical data and supports more efficient monitoring and decision-making.

GeoDefence: Interactive Threat Mapping and Monitoring System is a Python-based desktop application developed to provide an interactive platform for marking, monitoring, and managing location-based observations and potential threats. The system combines a desktop application interface with an interactive digital map, allowing users to directly identify geographical locations and record relevant information associated with them. The project is designed as a prototype to demonstrate how geographical information can be organized and visualized in a structured and user-friendly environment.

The application enables users to place markers on a map, assign a threat level such as High, Medium, or Low, and add descriptions or other relevant observations for each location. The use of different threat categories and color-coded markers helps users quickly distinguish between locations based on their assigned priority or significance.

Marked locations can also be repositioned by dragging the markers, while individual detections can be removed when they are no longer required. These features provide flexibility in managing information when location details or monitoring requirements change over time.

The system also includes a monitoring dashboard that provides a summary of the recorded detections. It displays the number of High, Medium, and Low threat locations, along with the total number of detections currently available on the map. This centralized overview makes it easier for users to monitor multiple locations and understand the overall distribution of recorded observations. The application also supports different map views, including normal map, terrain, and satellite views, providing users with different geographical perspectives depending on their requirements.

The current implementation of GeoDefence uses an interactive map interface integrated into a Python-based desktop application through PyQt6 and QWebEngineView. Leaflet is used to provide the interactive mapping functionality, including map navigation, markers, popups, and map layers, while HTML, CSS, and JavaScript are used to create and manage the visual interface and user interactions. Python manages the application environment and connects these technologies within a standalone desktop application. This combination provides a user-friendly system in which geographical information can be visualized and managed through direct interaction with the map.

In addition to online map functionality, the project explores the use of offline mapping technologies through locally stored map data and a tile-serving system. This provides a foundation for operating with geographical data in situations where continuous internet access may not always be available. The integration of such technologies demonstrates the potential for developing a more flexible mapping platform capable of supporting different operational environments.

The primary objective of GeoDefence is to create a functional prototype for location-based monitoring and threat visualization. The project demonstrates how interactive mapping technologies can be integrated with desktop applications to create a structured environment for recording, organizing, and managing geographically referenced information. It also provides practical experience in software development, user interface design, geospatial visualization, and the integration of multiple technologies within a single application.

Overall, GeoDefence serves as a foundation for the development of more advanced geographic monitoring systems in the future. The project can be further enhanced by implementing persistent and centralized databases, improved offline mapping capabilities, real-time data synchronization, authorized multi-user access, advanced search and filtering, historical data analysis, and AI-assisted analysis of geographical information. These potential improvements could expand the system from a prototype into a more comprehensive platform for structured location-based monitoring and decision support.

1.2 Background and Motivation

Geographical information plays an important role in understanding, analyzing, and monitoring events that occur across different locations. In many situations, information about important areas is collected through observations, reports, field personnel, surveys, sensors, or other available sources. This information may include the location of an observation, its level of importance, a description of the situation, and other relevant details. However, when such information is maintained only through written notes, lists, spreadsheets, or separate records, it can become difficult to understand the overall geographical situation and identify the relationship between different locations.

Managing a large number of location-based records without proper visualization can also create difficulties in monitoring and decision-making. Users may need to manually compare coordinates, search through multiple records, or refer to different sources of information to understand where a particular event or observation has occurred. As the number of recorded locations increases, organizing and updating this information can become increasingly challenging. A centralized system that presents geographical information visually can therefore make the monitoring process more structured, accessible, and easier to understand.

An interactive map provides a more practical and efficient way of handling such information. By placing important observations directly on a map, users can quickly understand where different locations are situated and view multiple points together within a single interface. A map-based system also allows users to navigate geographical areas, zoom into specific regions, and examine locations in greater detail. This makes geographical information easier to organize, update, visualize, and monitor.

Interactive mapping systems can also improve the way location-based information is classified and managed. For example, locations can be assigned different categories or priority levels depending on their significance. Using visual indicators such as different marker colors makes it easier for users to identify and distinguish between different types of recorded information. Additional details, descriptions, and observations can also be attached to individual locations, providing a more complete understanding of each recorded point.

GeoDefence: Interactive Threat Mapping and Monitoring System was developed based on this idea. The project provides an interactive map-based environment where users can mark geographical locations, add relevant information, assign a threat level, and manage the recorded points through a centralized interface. The system enables users to create location-based detections and classify them as High, Medium, or Low depending on the requirements of the monitoring process. The recorded points can be viewed directly on the map, allowing the user to understand their geographical distribution more clearly.

The project also includes features for managing the recorded information, such as repositioning markers, deleting unnecessary entries, switching between different map views, and viewing detection statistics through a monitoring dashboard. These features help provide a structured overview of the locations currently recorded in the system. The dashboard further simplifies monitoring by displaying the number of detections under different threat categories and the total number of recorded locations.

An important aspect of GeoDefence is that the system focuses on providing the user with direct control over the information rather than making autonomous decisions. The application acts as a tool for recording, organizing, and visualizing geographically referenced information. The user is responsible for adding observations, assigning threat levels, and managing the recorded data. This approach makes the system suitable as a prototype for exploring interactive mapping and location-based monitoring while maintaining human control over the information and decisions associated with it.

Overall, the motivation behind the development of GeoDefence is to demonstrate how modern mapping technologies and desktop applications can be combined to create a practical platform for geographical visualization and monitoring. The project provides a foundation for the future development of more advanced systems that could include features such as persistent data storage, offline maps, real-time synchronization, centralized databases, authorized user access, and additional analytical capabilities.

Motivation

The main motivation behind GeoDefence was to create a simple and practical way to convert location-based observations into a clear visual representation. Instead of depending on separate notes or manually maintaining lists of coordinates, the user can directly interact with a map and record information at the relevant geographical location. This approach makes it easier to understand the position of different observations and view them together within a single interface. It also reduces the difficulty of manually organizing geographical information from multiple sources.

Another motivation was to explore how different technologies can work together to create a useful real-world application. The project combines Python and PyQt6 for the desktop application environment with Leaflet and web technologies for interactive map visualization. HTML, CSS, and JavaScript are used to create an interactive and user-friendly interface within the desktop application. This combination helped in creating a system that is both technically interesting and easy for a user to interact with while providing practical experience in integrating multiple technologies.

The project was also motivated by the need for a flexible system. Situations and observations can change over time, so users should be able to update information easily. For this reason, GeoDefence allows marked locations to be edited, repositioned, reclassified, or deleted based on the user's requirements. The ability to change threat levels and update descriptions ensures that the information displayed on the map can be modified whenever new observations or requirements arise.

Another important motivation was to provide a centralized view of multiple geographical observations. By displaying different markers on a single map and categorizing them according to their assigned threat levels, the system can provide a clearer overview of the recorded information

Overall, the motivation behind GeoDefence is to build a foundation for a location-based monitoring system that makes geographical information easier to visualize, organize, and manage. The current project focuses on the core interactive mapping functionality and demonstrates how a map-based interface can improve the handling of location-related information. Future versions can be extended with features such as persistent databases, improved offline maps, real-time information sharing, authorized multi-user access, and AI-assisted analysis, allowing the system to become more advanced and capable of supporting larger monitoring requirements.

1.3 Scope and Limitations

The scope of GeoDefence: Interactive Threat Mapping and Monitoring System is to provide an interactive and user-friendly platform for visualizing and managing location-based observations on a geographical map. The project focuses on making geographical information easier to record, understand, organize, and manage through direct interaction with a digital map. By presenting location-based information visually, the system helps users view multiple observations together and understand their geographical distribution more clearly.

The application allows users to mark locations of interest directly on the map and associate relevant information with each marked point. Users can assign different threat levels, add descriptions, and update the information whenever required. This makes the system flexible and allows it to adapt to changes in the information being monitored. The color-coded classification of locations also provides a quick visual indication of their assigned threat level or priority.

The current version of GeoDefence includes features such as adding markers, editing existing locations, changing threat classifications, repositioning markers, deleting individual points, and clearing all marked locations when required. Multiple locations can be viewed together, providing a clearer geographical understanding of the information entered by the user. The system also includes a monitoring dashboard that displays the number of High, Medium, and Low threat detections along with the total number of recorded locations.

GeoDefence is developed as a Python-based desktop application using PyQt6, while HTML, CSS, JavaScript, and Leaflet are used to create the interactive mapping environment. The application also explores offline map functionality through locally stored geographical data and map tiles. The project demonstrates how desktop application development and web-based mapping technologies can be combined to build a practical geospatial monitoring system.

The scope of the current project is mainly focused on the development and demonstration of the core mapping and monitoring functionality. It provides a functional prototype in which users have direct control over creating, updating, classifying, and managing geographical information. The system does not make autonomous decisions but instead acts as a visual platform for organizing and monitoring user-provided data.
The project also provides a foundation for future development.

The existing system can be extended with features such as improved persistent data storage, centralized monitoring, authorized multi-user access, real-time updates, additional map layers, enhanced offline mapping capabilities, advanced search and filtering, historical data tracking, and intelligent data analysis. These future enhancements could expand GeoDefence into a more comprehensive and scalable geographical monitoring platform.

Limitations of the Project

The current version of GeoDefence is developed as a prototype and therefore has certain limitations. One of the main limitations is its dependence on an internet connection for loading and displaying map data. In areas with limited or no network connectivity, the map may not load properly or may not be available for use.

The current application is mainly designed for use as a local desktop system. It does not yet provide a centralized platform where multiple users can simultaneously access and manage the same information from different devices. Features such as real-time synchronization and centralized data sharing can be considered for future development.
Another limitation is related to data management and scalability. As the number of marked locations increases, a more structured and scalable storage system may be required to efficiently manage, search, and retrieve large amounts of geographical information.

The usefulness of the system also depends on the accuracy of the information entered by the user. Incorrect location marking or incomplete information may affect the overall quality and reliability of the visualized data.
In addition, the current project focuses primarily on the core functionality of interactive mapping and monitoring. Advanced features such as integration with external data sources, live location tracking, detailed geographical analysis, and large-scale deployment are beyond the scope of the present prototype.

Despite these limitations, GeoDefence successfully demonstrates the core concept of interactive geographical monitoring and provides a strong base for future improvements. The current prototype can be further developed into a more advanced, scalable, and feature-rich geospatial monitoring system.

1.4 Problem Statement

In many monitoring and field-based situations, important information is connected to specific geographical locations. When multiple observations or locations require attention, managing this information through separate notes, written reports, or lists of coordinates can become difficult. Such methods may not provide a clear overall view of where important locations are situated or how multiple recorded points are distributed geographically.

There is a need for a simple and interactive system that allows users to visualize location-based information directly on a map and manage it in an organized manner. Users should be able to easily identify a location, mark it on the map, add relevant information, classify its level of importance, and update the information whenever required.

GeoDefence: Interactive Threat Mapping and Monitoring System addresses this problem by providing a map-based desktop application for recording and managing location-based observations. The system allows users to place markers on an interactive map, assign threat levels, add descriptions, edit existing information, reposition markers, and remove locations when they are no longer required.

The main problem addressed by the project can therefore be summarized as the need for a simple, interactive, and organized platform for visualizing and managing geographically referenced observations and potential threat locations. GeoDefence aims to bring location information and user interaction together in a single interface, making it easier to monitor and understand multiple locations from a geographical perspective.
The project focuses on developing a functional prototype that provides the foundation for a more advanced geospatial monitoring system in the future.

1.5 Motivation

The motivation behind developing a map using Python is to create a reliable and efficient geographic information system that can support various defense-related applications. Accurate and detailed geographical information plays an important role in areas such as navigation, route planning, terrain analysis, monitoring, and situational awareness. A well-designed mapping system can help users understand geographical environments more effectively and make informed decisions based on location-based data.

Python was selected for this project because of its flexibility, extensive libraries, and ability to handle large amounts of geographical and spatial data. Using Python, the system can integrate different map layers such as roads, state and district boundaries, rivers, terrain, landmarks, and other relevant geographical features. It can also provide interactive features such as zooming, panning, searching locations, displaying coordinates, and adding customized markers.

The project is particularly motivated by the need to explore how modern mapping technologies can be adapted for defense-oriented applications. A detailed map can serve as a foundation for understanding terrain, identifying important geographical features, supporting planning activities, and improving overall situational awareness. The system can also be further expanded in the future by integrating additional datasets, analytical tools, and intelligent features.

Overall, this project provides an opportunity to combine Python programming, geographic data processing, GIS concepts, and user-interface development into a practical real-world application. It demonstrates how software-based mapping solutions can contribute to the development of reliable and efficient technologies for defense and other critical environments.

2. About Company

2.1 About ATWIC R&D

ATWIC Research & Development Pvt. Ltd. (ATWIC R&D) is a deep-tech research and engineering company headquartered in Kakkanad, Kochi, Kerala. The company focuses on developing advanced sensing technologies, electronic systems, and integrated engineering solutions for applications across defense, aerospace, oceanography, environmental sciences, scientific research, and industry.

Founded by Rahul Sivan, ATWIC R&D brings together multidisciplinary expertise in hardware design, firmware development, mechanical engineering, scientific research, signal processing, and AI-based implementation. Its approach combines scientific principles with practical engineering to transform complex physical phenomena into measurable and usable technological solutions.

Core Areas of Expertise

The company has expertise across several technical domains:

- Hardware Design – Analog, digital, and power electronics.
- Firmware Development – Device drivers, digital signal processing (DSP), and AI implementation.
- Mechanical Engineering – MEMS, packaging, and thermal design.
- Scientific Research – Optics, magnetism, radiation, and particle physics.
- Advanced Sensors – Development of precision sensing technologies for different environments.
- Integrated Systems – Combining sensors, processing systems, communication technologies, and analytics.
- Custom Development – Designing and modifying solutions according to specific technical and operational requirements.

Major Application Areas

ATWIC R&D works across several strategic and scientific application areas. Its defense and aerospace work includes sensing and inertial systems, while its maritime activities involve underwater acoustic technologies and marine sensing. The company also highlights border and underground monitoring technologies, scientific instrumentation, and environmental applications.

Its technology portfolio includes areas such as gyroscopic systems, magnetometers, LIDAR, acoustic sensing, inertial navigation, communication systems, data acquisition, optical systems, and magnetism detection.

Research & Development Approach

A major strength of ATWIC R&D is its multidisciplinary approach. Rather than focusing on a single technology, the company combines physics, electronics, software, mechanical engineering, sensing, and signal processing to develop complete technological solutions.

The company also provides custom engineering, rapid prototyping, system integration, testing, and product development, allowing research concepts to progress toward practical applications.

Company Vision

ATWIC R&D's broader vision is to establish itself as a center of excellence in sensing technology, developing innovative and reliable solutions for strategic, scientific, and industrial applications. Its work reflects an emphasis on converting advanced scientific concepts into practical systems that can address real-world engineering challenges.

3. Working Domain Or Technologies Used

3.1 Python

Python is the primary programming language used in the development of the GeoDefence application. It is responsible for initializing the application, creating the main program structure, and managing the desktop application environment. In this project, Python is used together with the PyQt6 framework to create the main application window and load the interactive mapping interface.

Python also acts as the connecting layer between the desktop application and the web-based map components. It provides the environment in which the application runs and allows different technologies to be integrated into a single system. Its simple syntax, extensive library support, and ability to integrate with graphical user interface frameworks make it suitable for developing this type of prototype.

3.2 PyQt6

PyQt6 is a Python framework used to develop the Graphical User Interface (GUI) of the GeoDefence system. It provides the components required to create a standalone desktop application, including the main application window and the graphical environment in which the mapping system operates. In this project, QApplication is used to manage the application's execution, while QMainWindow provides the main window structure.

The use of PyQt6 allows the mapping application to operate as a dedicated desktop program rather than requiring the user to open the map directly through a normal web browser. It provides a structured application environment and makes it possible to combine Python programming with modern web-based technologies.

3.3 Qt WebEngine / QWebEngineView

QWebEngineView is used to display the HTML-based map interface inside the PyQt6 application. Since Leaflet works using HTML, CSS and JavaScript, WebEngine acts like a built-in browser that renders all of it. This is what allows your Python desktop application to show the interactive map.

3.4 HTML

HTML, or HyperText Markup Language, is used to provide the basic structure of the interactive mapping interface. It defines the different elements that are displayed to the user, including the map container, monitoring dashboard, control buttons, detection forms, popups, labels, and instructions.

In this project, HTML acts as the foundation of the visual interface. It organizes the different components of the application and provides the structure required for CSS styling and JavaScript functionality. Without HTML, the various interface elements used for interaction and displaying information would not have a structured layout.

3.5 CSS

CSS, or Cascading Style Sheets, is used to control the visual appearance and layout of the GeoDefence interface. It is responsible for designing the monitoring dashboard, buttons, threat indicators, popup forms, fonts, spacing, borders, and other visual elements within the application.

The project uses CSS to create a dark and structured interface suitable for a monitoring system. Different colors are also used to visually represent threat levels and make the information easier to understand. CSS improves the overall user experience by ensuring that the application interface is organized, readable, and visually consistent.

3.6 JavaScript

JavaScript is responsible for most of the interactive functionality within the mapping interface. It controls how the application responds when the user interacts with the map, buttons, markers, and detection forms. The core functionality for creating and managing detections is implemented using JavaScript.

In GeoDefence, JavaScript is used to add new detection markers when a user clicks on the map, generate unique detection IDs, save detection information, assign threat levels, change marker colors, delete markers, and update the monitoring dashboard.

3.7 Leaflet.js

Leaflet.js is the primary JavaScript mapping library used to create the interactive map in the GeoDefence system. It provides the functionality required to display geographical maps and allows users to interact with them through features such as zooming, panning, clicking, markers, popups, and map layers.

In this project, Leaflet.js is responsible for the core geographical visualization functionality. It allows users to click on a location and create a detection marker, display information through popups, move markers to different locations, and switch between multiple map views. Leaflet provides a lightweight and flexible solution for developing interactive map-based applications.

3.8 OpenStreetMap

OpenStreetMap is used as the primary or normal map layer in the application. It provides geographical map tiles that display information such as roads, cities, locations, boundaries, and other geographical features. These tiles are loaded and displayed through the Leaflet.js mapping library.

In GeoDefence, OpenStreetMap provides the standard geographical view that users can use for navigation and location monitoring. It serves as the default map layer when the application is opened and provides a clear representation of the geographical area being viewed.

3.9 OpenTopoMap

OpenTopoMap is used to provide a terrain-based view of the geographical area. Unlike a standard map, a terrain map provides greater emphasis on geographical and topographic features, helping users understand the physical characteristics of an area.

In the GeoDefence application, OpenTopoMap is included as an alternative map layer that users can select through the layer control. This gives the user the flexibility to switch between a standard geographical view and a terrain-focused view depending on the type of information they want to examine.

OpenTopoMap provides the terrain view of the map. It is useful when geographical elevation and terrain characteristics are important for understanding an area. Your project includes it as an alternative layer that users can switch to from the map controls.

3.10 Esri World Imagery

Esri World Imagery is used to provide a satellite imagery view of geographical locations. This layer displays aerial imagery and provides a more realistic visual representation of the land and surrounding environment compared with standard map representations.

In GeoDefence, the satellite layer is available as one of the alternative map views. Users can switch between the Normal, Terrain, and Satellite layers according to their requirements. The availability of multiple map views improves the flexibility of the application and allows geographical information to be viewed from different perspectives.

3.11 Leaflet DivIcon

Leaflet's L.divIcon() functionality is used to create custom markers for displaying detection locations. Instead of using the standard Leaflet marker icon, the project creates circular markers using HTML and CSS. This allows the appearance of each marker to be dynamically controlled.

In GeoDefence, the marker color changes according to the threat level assigned to a detection. A red marker represents a High threat, an orange marker represents a Medium threat, and a green marker represents a Low threat. This color-based approach makes it easier for users to quickly identify and distinguish between different classifications directly on the map.

3.12 Web-based Map Tiles

The application uses web-based map tile services to display geographical information. A map is divided into multiple small image tiles, and the appropriate tiles are loaded based on the user's current geographical position and zoom level. This allows large geographical areas to be displayed efficiently without loading the entire map as a single image.

In GeoDefence, different tile services are used to provide the Normal, Terrain, and Satellite map layers. Leaflet.js dynamically requests and displays the required map tiles as the user moves or zooms across the map. This approach provides smooth map navigation and allows different geographical views to be integrated into the same application.

4.About Team And Reporting Manager

Team and Guidance

During my internship, I had the opportunity to work under Rahul, the CEO of the company, who provided overall leadership and direction during my internship.

I was part of the AI Development Team, working under Sreerag, the Head AI Developer and my Reporting Manager. He was responsible for guiding me throughout the project, explaining the technical requirements, reviewing my progress, and providing valuable feedback whenever needed.

Working under his supervision gave me practical exposure to developing a real-world application and helped me strengthen my technical, problem-solving, and project development skills. The guidance and support from both Rahul and Sreerag contributed significantly to the successful completion of my internship project.

Team Summary

I worked under Rahul, the CEO of the company, with Sreerag, the Head AI Developer and my Reporting Manager, as my primary technical guide. As part of the AI Development Team, I received guidance on project requirements, implementation, and development. Their support helped me gain practical experience in working on a real-world technical project and improve my problem-solving and development skills.

5.Challenges Faced In The Internship

During the internship, I faced several challenges while developing the Defense Map System. One of the main challenges was understanding how different technologies such as Python, PyQt6, QWebEngineView, HTML, CSS, JavaScript, and Leaflet.js could be integrated into a single application. Since the project involved an interactive map, handling map layers, markers, popups, and user interactions required careful implementation and testing.

Another challenge was working with different map tile services such as OpenStreetMap, OpenTopoMap, and Esri satellite imagery. Ensuring that the different map views loaded correctly and switching between them worked smoothly required troubleshooting. Designing a user-friendly dashboard and implementing features such as threat-level markers, detection counts, marker movement, and deletion also required attention to detail.

Understanding project requirements and converting them into practical features was another learning challenge. However, with guidance from my Reporting Manager, Sreerag, and continuous testing and debugging, I was able to overcome these challenges and successfully develop the project.

6.Objectives of the Work

The main objective of the internship work was to develop an interactive Defense Map System that could be used for geographical monitoring and detection marking. The project focused on combining mapping technologies with a desktop application to create a simple and user-friendly monitoring interface.

The key objectives were:

1. To develop an interactive digital map that allows users to view and navigate different geographical locations.
2. To integrate multiple map views, including Normal, Terrain, and Satellite views, to provide different perspectives of the monitored area.
3. To enable detection marking, allowing users to click on the map and create location-based detection points.
4. To categorize detections based on threat level using High, Medium, and Low classifications with different colored markers.
5. To provide detection information, including a detection ID, name, threat level, description, and geographical location.
6. To develop a monitoring dashboard that displays the total number of detections and their respective threat-level counts.
7. To create a desktop-based interface using Python and PyQt6, making the map system accessible as a standalone application.
8. To gain practical experience in integrating Python with web technologies such as HTML, CSS, JavaScript, and Leaflet.js while developing a real-world application.

7.Github Link

https://github.com/felixsajeev2006-blip/GeoDefence-Interactive-Threat-Mapping-and-Monitoring-System

8.Output

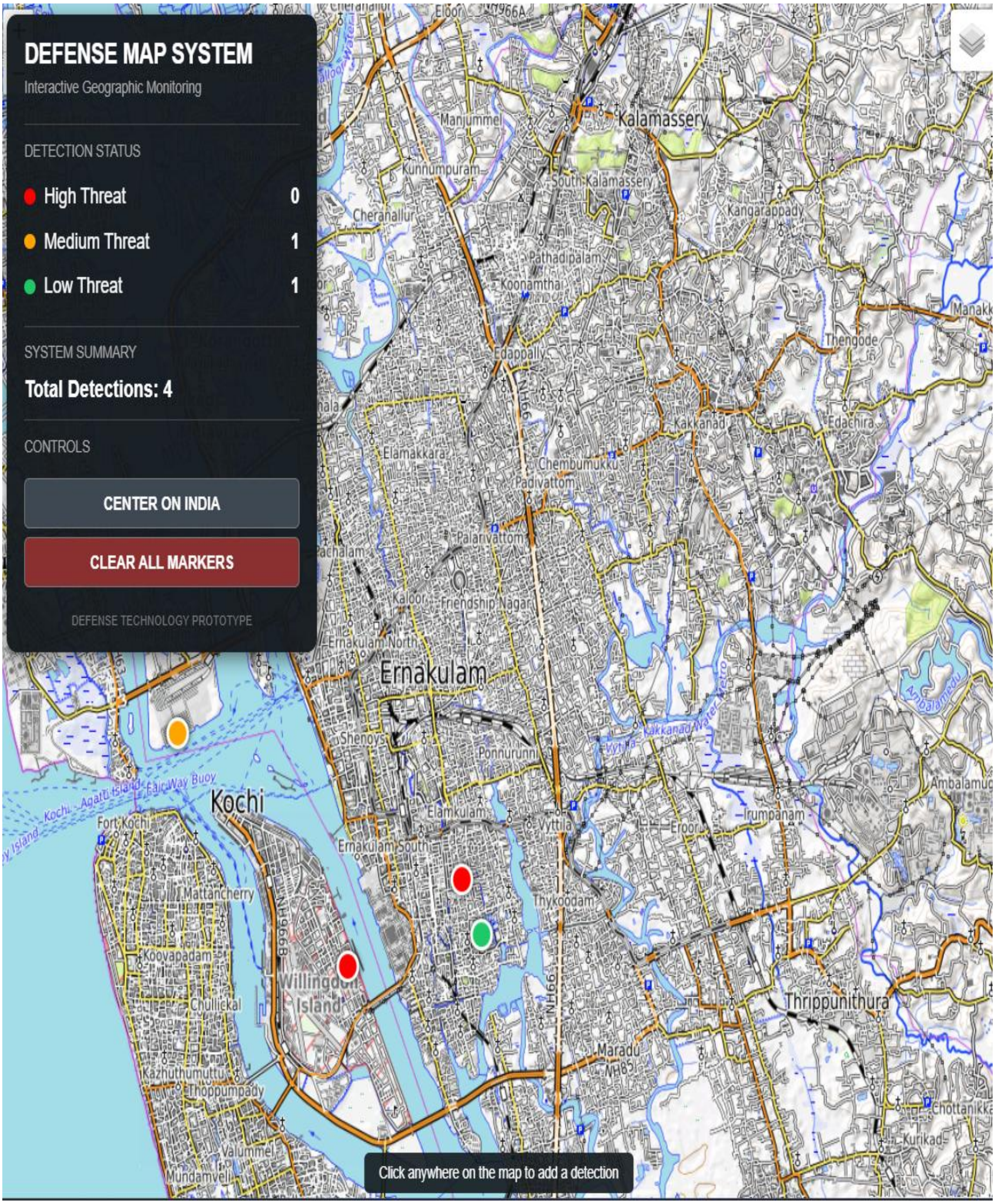

Figure 1: Zoomed-in Terrain Tile View Displaying the Marked Location

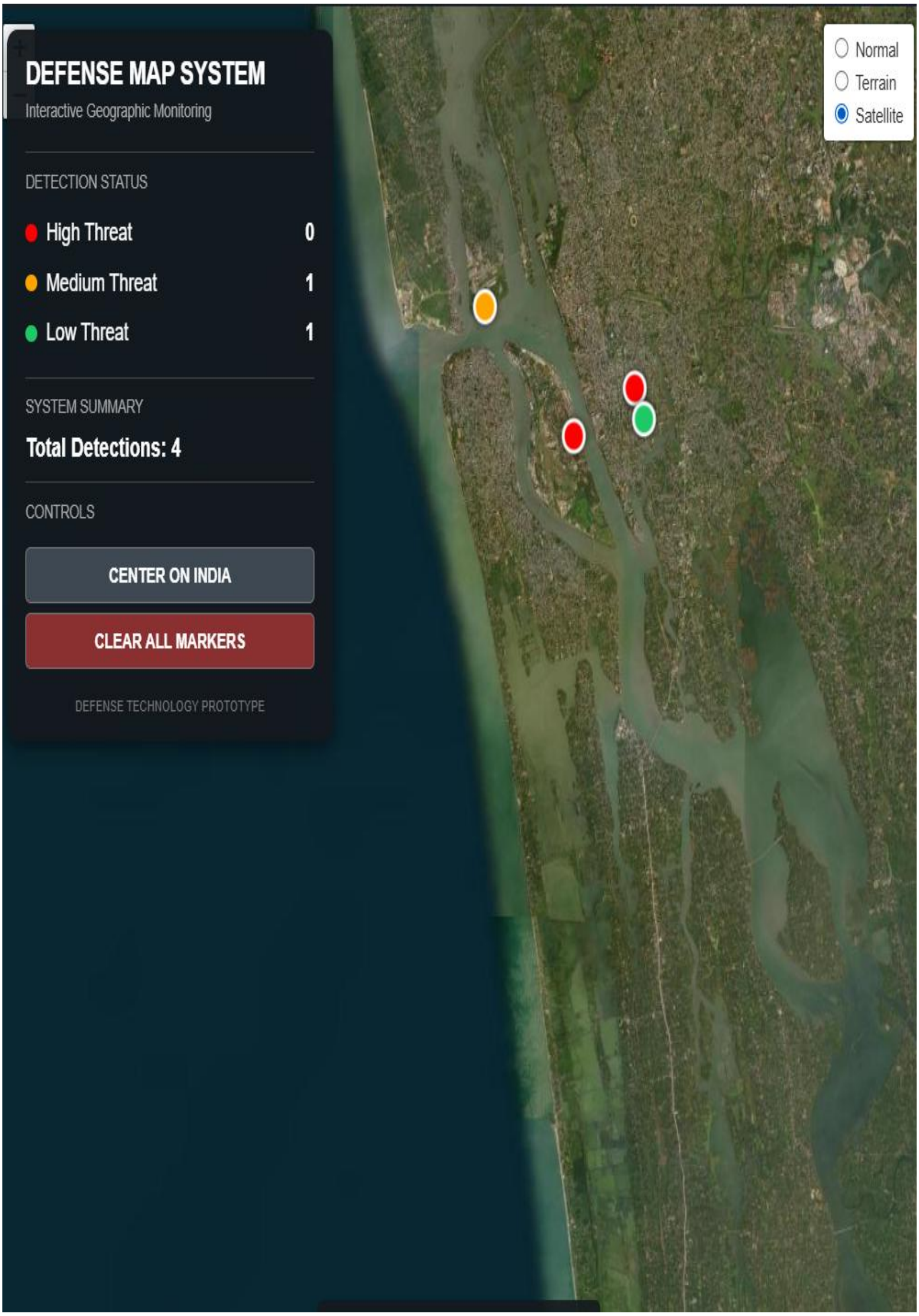

Figure 2: Marked Location in Satellite View

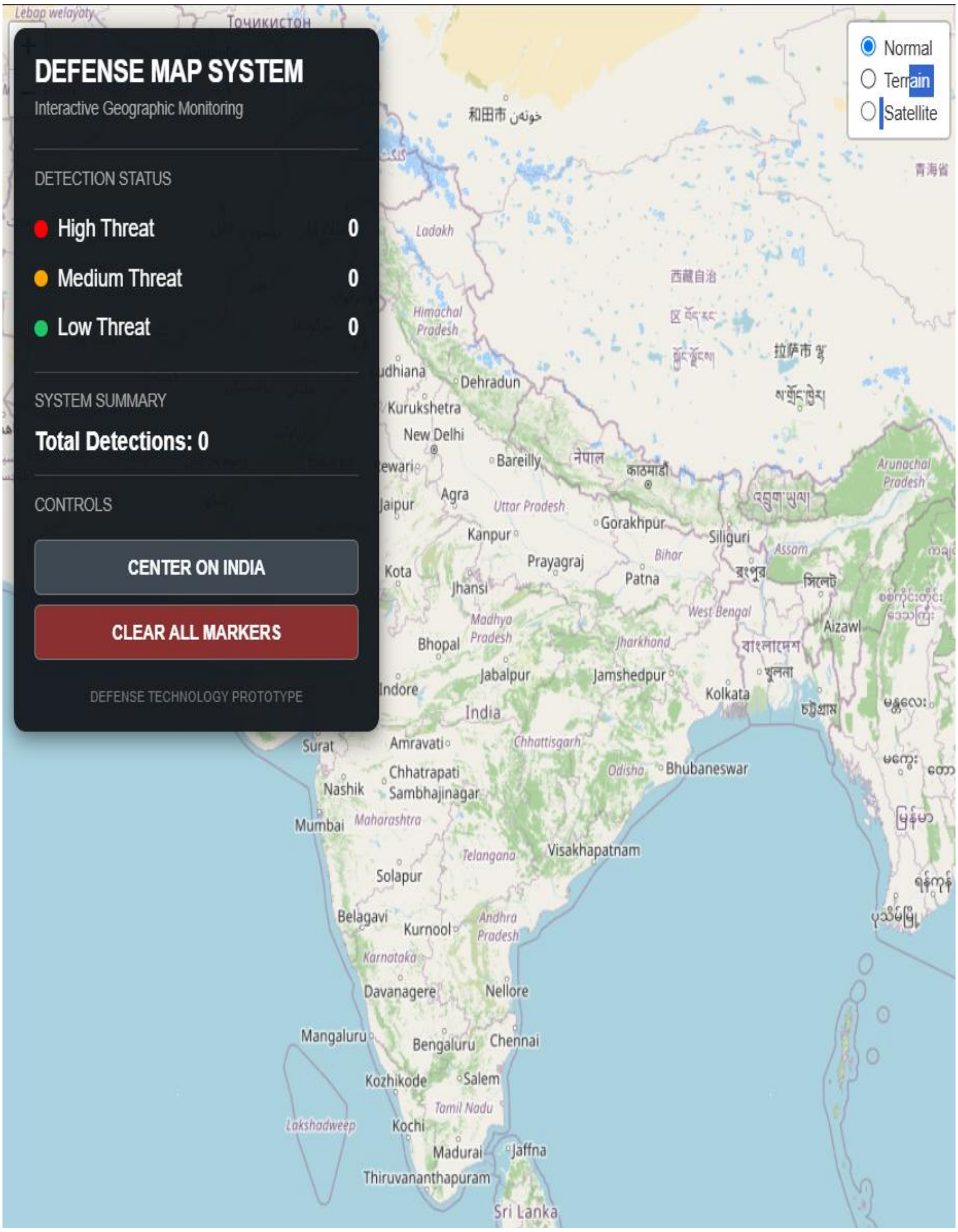

Figure 3: OpenMap View

Figure 4: Terrain View at a Zoomed-Out Level

9.Conclusion

The GeoDefence – Interactive Threat Mapping and Monitoring System successfully demonstrates the development of a geographic monitoring application using modern software and mapping technologies. The project combines Python, PyQt6, Leaflet.js, HTML, CSS, JavaScript, Flask, SQLite, and MBTiles to create an interactive platform for visualizing and managing location-based detection information.

Through this project, users can mark locations, classify detections based on threat levels, add relevant information, and monitor detection statistics through a centralized dashboard. The integration of offline map support also provides a foundation for using geographical data in environments where reliable internet connectivity may not always be available.

Overall, the project provided valuable practical experience in software development, geospatial visualization, user interface design, offline map handling, and integration of multiple technologies. It also established a foundation that can be further enhanced with persistent data storage, advanced filtering, authentication, and additional geospatial analysis capabilities.

10.References

1. Python Software Foundation, Python Documentation. Available at: https://docs.python.org/3/

2. Qt, Qt for Python (PySide/PyQt-related Qt documentation). Available at: https://doc.qt.io/qt-6/

3. Leaflet, Leaflet JavaScript Library Documentation. Available at: https://leafletjs.com/reference.html

4. OpenStreetMap Foundation, OpenStreetMap Documentation. Available at: https://wiki.openstreetmap.org/

5. OpenTopoMap, OpenTopoMap – Topographic Maps. Available at: https://opentopomap.org/

6. Esri, ArcGIS World Imagery and Map Services Documentation. Available at: https://developers.arcgis.com/

7. Flask, Flask Documentation. Available at: https://flask.palletsprojects.com/

8. SQLite, SQLite Documentation. Available at: https://www.sqlite.org/docs.html

9. Mapbox, MBTiles Specification. Available at: https://github.com/mapbox/mbtiles-spec

10. GeoDefence – Interactive Threat Mapping and Monitoring System, GitHub Repository. Available at: https://github.com/felixsajeev2006-blip/GeoDefence-Interactive-Threat-Mapping-and-Monitoring-System